

Problem 9

Problem 9

Problem 1 Compute $f'(x)$.

(a) $f(x) = x \ln(x)$

Explanation. $f'(x) = x \cdot \frac{1}{x} + \ln(x) = 1 + \ln(x)$

(b) $f(x) = \sin(x) \left(\ln(\sec(x) + 1) \right)$

Explanation. $f'(x) = \sin(x) \frac{\sec(x) \tan(x)}{\sec(x) + 1} + \cos(x) \left(\ln(\sec(x) + 1) \right)$

(c) $f(x) = 2^x \sqrt{\ln(5x + 7)}$

Explanation. $f'(x) = \frac{2^x}{2\sqrt{\ln(5x + 7)}} \cdot \frac{5}{5x + 7} + 2^x \ln(2) \sqrt{\ln(5x + 7)}$